



github.com/sailuh

SAILUH

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Sponsored by: Carlos Paradis and Rick Kazman

What is SAILUH?

Software Analytics Insight Lab

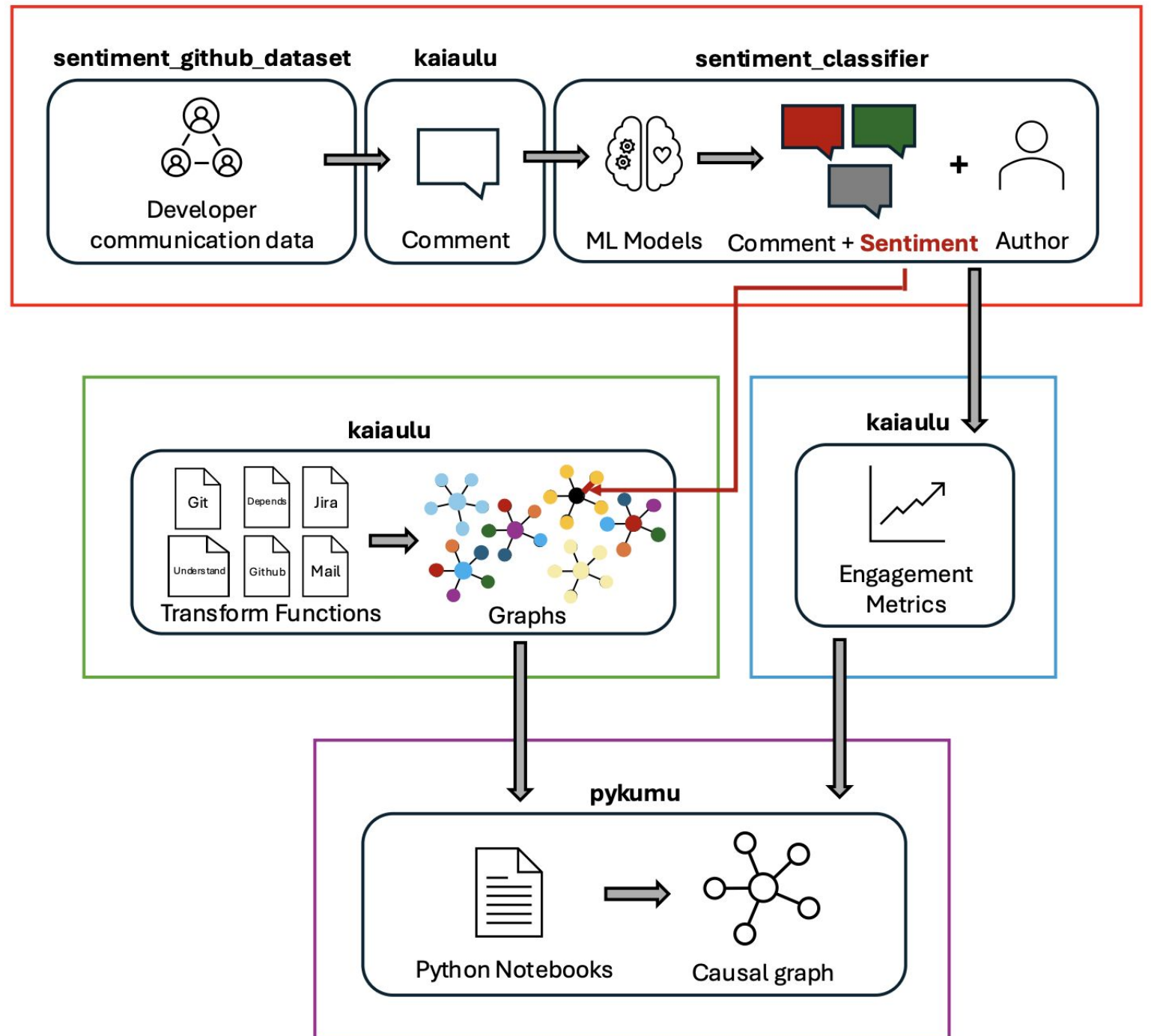
A GitHub organization for a UH Mānoa research lab that builds tools to analyze software projects and developer communities.

Problem Statement

A subset of the SAILUH ecosystem does not yet provide interoperability among its tools to identify how developers' sentiment affects project health.

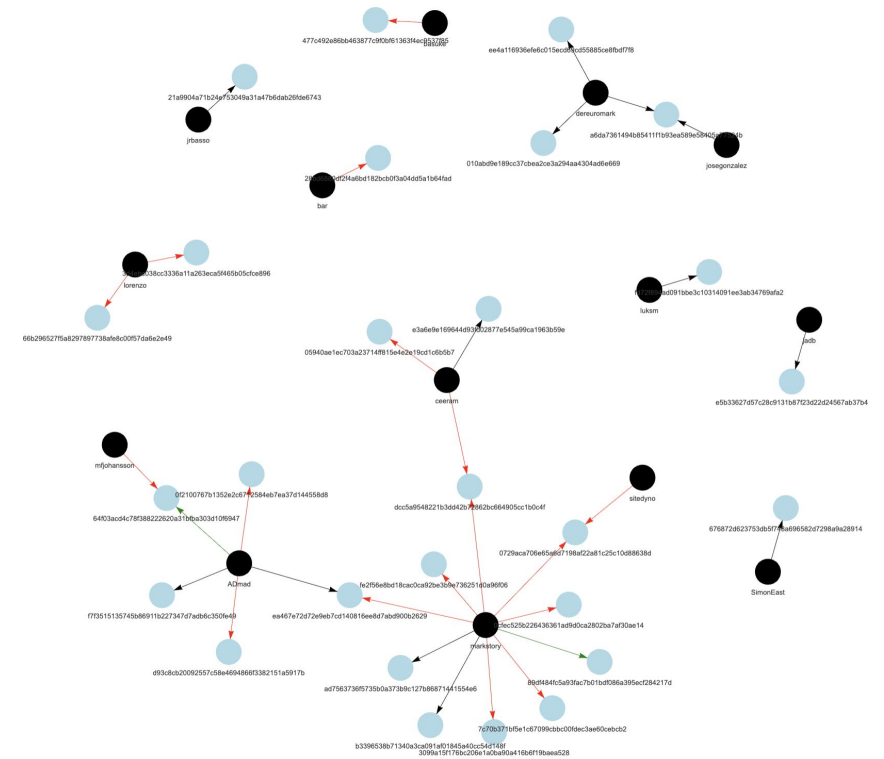
- Sentiment Classifier
- Engagement Trends
- Network Refactoring
- Causal Analysis

Technical Solution Overview



Major Accomplishments

- **Sentiment Classifier:** Built an end-to-end pipeline that produces sentiment-labeled communication comments compatible with Kaiaulu.
- **New Metrics:** Implemented new metrics that measure engagement, sentiment, and productivity among developers in a project.
- **Network Refactoring:** Introduced a standardized S3 graph model with typed constructors and dispatch methods for extensible downstream analysis.
- **Causal Analysis:** Implemented Pykumu API package, a Python interface to Tetrad's causal analysis library.



Quality Assurance Metrics

- Submitted pull requests with new features and fixes
- Incorporated sponsor feedback through PR and Issue discussions before a merge
- Maintain a “Project Overview” issue tab
 - Project Scope
 - Requirements
 - Milestones
 - Weekly notes
 - Next steps

Implementing Productivity Metrics #380

usradam wants to merge 14 commits into master from 377-implement-productivity-metrics

Conversation 32 · Commits 14 · Checks 2 · Files changed 44

+550 -89

usradam commented on Mar 18 · edited · Collaborator

Purpose

Coming off [the engagement metrics](#), we now want to measure developer productivity through commits and lines of code (LOC).

Per Rick's email:

"Quitting is simply the low end of the engagement scale and rage is the high end of the sentiment/emotion scale. But in this study we are seeking a more nuanced understanding of the relationships. The hypothesis would be that lower sentiment/emotion scores will correlate with lower engagement and lower productivity."

Process

- Have the productivity metrics for commits and churn be implemented as two functions called 1) `productivity_author_commits` and 2) `productivity_author_churn`
- Understand the input and output of these functions
- Create the function signatures, ensuring relevant parameters are included
- Both functions take into account the 90-day rolling window for its aggregation as we have looked at before with the engagement metrics

Implementing Commits Productivity

For commits we can use `commit_hash` from the `parse_gitlog` data table to get the unique commits per `author_name_email`. I can then get a simple summation of the overall number of commits. The resulting output would

Reviewers

- carlosparadis
- jseto808
- +1 more reviewer
- phuong808

Still in progress? Learn about draft PRs

Assignees

- usradam

Labels

None yet

Projects

- ics496-spring26-causal-sentiment-r...
- Status: Code Ready for Arc...
- +5 more

Milestone

- ics-496-spring26-engagement-metrics

Development

Migrate Workflow from R Kumu to PyKumu (Python) #4

phuong808 opened on Feb 2 · edited by phuong808

Edits · Collaborator

Project Overview

The purpose of this project is to migrate an existing R analysis notebook to Python, verify that the Python implementation produces equivalent results, refactor the workflow into a modular API, and integrate the Kaiaulu data processing pipeline to support a complete and automated data analysis process.

Milestone 1

- ☒ Fully migrate the existing R notebook into a Python notebook and confirm that both implementations produce identical outputs.

Milestone 2

- ☐ Refactor the Python notebook into reusable Python modules and expose its core functionality through an API.

Milestone 3

- ☐ Integrate the Kaiaulu data file into the Python workflow by parsing and preprocessing the data prior to performing causal analysis.

Task List

Assignees

- phuong808

Assign to Agent

Labels

No labels

Type

No type

Projects

- ics496-spring26-causal-sentiment-r...
- Status: Task ready for codi...

Milestone

- Milestone 1
- Past due by 1mo 4d, 33% complete

Relationships

None yet

Development

Project Management and Delivery

Implement GitHub Commit Comments Downloader Function #378

[Open](#)

#381



splimon opened on Mar 11 · edited by splimon

Edits

Collaborator

Purpose

Kaiaulu currently has the function `github_api_project_commits` to download commits from the GitHub API, but no equivalent function for downloading commit comments.

This new function is needed to transfer sentiment labels attached to commit comments from the [GHTorrent database](#). To map these labels to Kaiaulu-downloaded data, the comment text must be retrieved through Kaiaulu's GitHub API pipeline rather than sourced directly from the static GHTorrent database. The bridge between the sentiment label and the downloaded comment is the **commit comment ID**.

This was identified and discussed in [sailuh/sentiment_github_dataset#1](#).

Process

Implement a new function called `github_api_project_commit_comments` in `R/github.R` by:

sailuh / kaiaulu

Type to search

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Implement GitHub Commit Comments Downloader Function #381

Unable to merge

[Open](#)splimon wants to merge 8 commits into `sailuh:master` from `splimon:378-github-commit-comments-downloader-function`

Conversation 3



Commits 8



Checks 2



Files changed 90

Commits on Mar 19, 2026

feat: download and parse Github commit comments

splimon committed on Mar 19 · ✓ 2 / 2

958cc3

Commits on Mar 20, 2026

feat: add commit comments download workflow and fix gh_next() page bug

46324d

Define Consistent Transform Interfaces and Graph Abstractions Across Kaiaulu Pipelines

#361 · jseto808 opened on Feb 2 · ics-496-spring26...

1

58



Reflections

- We learned how to contribute to open-source software while aligning with SAILUH's architecture
- We discovered how to effectively extend SAILUH's capabilities through mutually agreed upon specifications from our sponsors
- Wrote maintainable code for future contributors
- The project improved our technical skills by working on a large-scale software project, utilizing industry standard developer tools like RStudio, GitHub, and Git.

Next Steps

SAILUH is ever-evolving...

Sentiment Classifier

Clean up the sentiment classifier codebase to ensure the full pipeline is compatible with the data format produced by dataset preparation.

Metrics

Expand the connection of the data pipelines to the metrics for the R notebooks.

Network Refactoring

Create new R notebooks that document the additional functionality enabled by transform standardization.

Causal Analysis

Apply the pykumu pipeline to an engagement metrics dataset to produce a causal graph.

Acknowledgements

Carlos Paradis

Data Scientist at KBR, Inc.

Ph.D. in Computer Science from
UH Manoa



Rick Kazman

Professor of Information
Technology Management at
Shidler College of Business

